SUPPORTING ADULT GUIDANCE

Conversion graphs as models

Straight Lines and Connected Representations. For any supporting adult — teacher, practitioner, parent or carer. **You do not need to be a mathematics specialist to support this lesson.**

What this lesson does

The learner uses a straight line as a **conversion graph** — a model that translates one quantity into another. The emphasis is on **reading the graph in both directions** and interpreting the **gradient as the conversion rate**, including the important case of a line that does **not** pass through the origin.

Driving Question: *How can a line translate one quantity into another?*

The mathematics, in plain terms

A **conversion graph** is a straight line that turns one quantity into another. To use it, you read **to the line**: go up from the value you know until you meet the line, then across to read the answer off the other axis — and it works in **either direction** (across-and-down converts back). The **gradient is the conversion rate**: for miles to kilometres the line climbs about 1.6 km for every mile, so 5 miles reads across to 8 km. For pounds to euros at £1 = €1.20, £40 reads across to €48, and €30 reads back to £25. The case worth dwelling on is temperature: **°C to °F** follows F = 1.8C + 32, so the line crosses the °F axis at **32, not 0** — 0°C is 32°F. This is exactly why you read to the line and do not simply multiply: 10°C is not 18°F (that forgets the +32), it is 50°F. Conversion graphs are everywhere useful — changing money on holiday, following a foreign recipe, reading distances or temperatures in the 'other' units. Once the line is drawn, every conversion is a quick up-and-across or across-and-down, in whichever direction you need.

Read to the line, both ways

The heart of this lesson is a reliable habit: read **to the line**, then across — and remember it works in both directions from the same line. Naming the two common slips helps: reading off the wrong axis, and 'just multiplying' without the graph, which quietly goes wrong whenever the line does not start at zero. The °C-to-°F example is the one to protect, because it is the clearest case where the line does not pass through the origin; seeing that 0°C sits at 32°F, not 0, makes the whole idea of 'read to the line' land. It helps to keep the gradient in view as the conversion rate — a steeper line simply means more of one unit per unit of the other. Grounding the reading in a genuine need (holiday money, a recipe, a road sign abroad) keeps the graph from being an abstract exercise and shows why a single line, read both ways, is such a handy tool.

The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

Cornerstone	Its role here	Invitational timing
Connection	Conversion Graph Translator: read a value up to the line and across to the other axis.	~12 min

Movement	Two-way reader: drag a point along the line; guides drop to both axes, reading the conversion both ways.	~12 min
Reflection	Read it right?: spot the slips — reading the wrong axis, or just multiplying when the line does not start at zero.	~10 min
Creativity	Holiday money: use a pounds-to-euros graph to convert both ways for real questions.	~6 min
Rest	A pause after seeing one line translate endlessly between two quantities.	~3 min
Nutrition	Contextual (Mode A): money, recipes and travel — a conversion graph is a look-up you can read both ways.	~2 min

Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

Reading off the wrong axis

Repair: “Which quantity is on each axis? Start from the value you know, go to the line, then read the OTHER axis.”

Just multiplying when the line does not start at zero

Repair: “Does this line pass through (0, 0)? For °C to °F it does not — 0°C is 32°F, so you must read to the line.”

Forgetting the graph converts both ways

Repair: “You went up-and-across to convert one way. Can you go across-and-down to convert back? Same line.”

Not reading all the way to the line

Repair: “Follow the dashed guide right up to the line before you read across — stopping short gives the wrong value.”

Confusing the two units when stating the answer

Repair: “Say the answer with its unit: is this 8 km or 8 miles? Which axis did you read it from?”

Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

Please hold this line

Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

Reconnection Routes

If an earlier idea is blocking progress, the lesson’s Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: reading coordinates from a graph, reading a value up to a line, and understanding the gradient as a rate. A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

Supporting through questions: an example

Keep the read-to-the-line habit and the units with the learner:

- “Which value do you know, and which axis is it on?”
- “Go up to the line — now read across. What is the answer, and in what units?”
- “Can you use the same line to convert back the other way?”
- “Does this line pass through (0, 0)? If not, why can’t you just multiply?”

If they stall, that is the moment for a Reconnection Route or the lesson’s prompt and hint — not for the answer.

Questions you might have

“Why can’t I just multiply by the rate?”
For conversions through the origin (miles to km, £ to €) multiplying works fine. But when the line does not start at zero — like °C to °F, which adds 32 — multiplying alone is wrong. Reading to the line always works, whichever kind of conversion it is.

“How do I convert the other way?”
Use the same line. To go the first way you read up-and-across; to convert back you read across-and-down. One line does both directions — that is what makes a conversion graph so handy.

“What does the gradient tell me?”
It is the conversion rate: how many of one unit you get for each of the other. A steeper line means a bigger rate — for example, more kilometres per mile than euros per pound.

Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

Recognising progress

Progress is described as Emerging → Developing → Secure → Mastering across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

Where they are	What it can look like
Emerging	Beginning to engage; names some features and makes a start with support.

Developing	Carries out the method with prompts; can explain part of the reasoning.
Secure	Works through independently and explains why each step is true.
Mastering	Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else.

A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.

NEO by Nudge Education · nudgeeducation.online · Curriculum by Gerry Docherty. Supporting Adult Guidance v0.1.